

KEY FIGURE

Alexander John Forsyth
(1768–1843)

Alexander Forsyth graduated from King's College, Aberdeen, Scotland, in 1786, and in 1791, he was licensed as a minister in Belhelvie, Aberdeenshire. He was a game shooter as well as an amateur chemist and mechanic. His frustration with the flintlock's weaknesses spurred him to devise a better ignition system.



Joe Manton designed the “tube-lock”—in this, he placed the fulminate in a thin copper tube, which was inserted into a vent on one side of the barrel and struck with a hammer. Other systems included the “pellet-lock” and Edward Maynard's tape primer. The tape primer had the fulminate in a series of “caps” in a long tape and was popular in the US for a while. Even in recent times this was the “ammunition” for toy cap guns.

“... one of the most ingenious... one of the most useful inventions in modern times...”

ATTRIBUTED TO COMMITTEE OF PATENTS ON JOSHUA SHAW'S CLAIM (FEBRUARY 1846)

THE PERCUSSION CAP

The breakthrough, however, was made in 1822 by Joshua Shaw, an English artist. He designed a tiny copper cup, put fulminate in it, and held it in place with a drop of varnish. Shaw placed this cuplike cap on a hollow plug, or nipple, screwed into the breech of a gun, ready to be struck by the hammer. Striking the cap ignited the primer, producing a flash that was relayed to the propellant via a vent in the barrel.

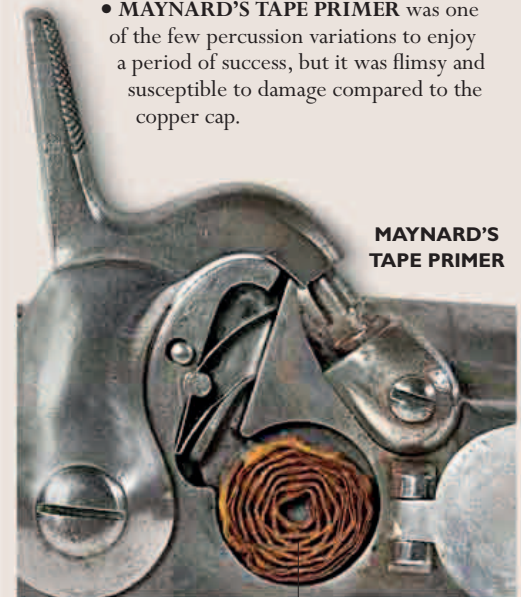
As the percussion system evolved, ultimately resulting in the percussion cap, guns were transformed by having a means of ignition that was reliable and easy to use. Reloading times for these guns decreased dramatically. Rifles employing percussion caps were common in the Crimean War (1853–56). An important battle in this war was the Battle of Balaclava, in which a small number of British troops armed with percussion rifles stood their ground against a Russian cavalry onslaught, firing at the larger force in a volley. The percussion rifles were precise and reliable, and they could be reloaded quickly, which allowed the British forces to repel the Russians. Percussion weapons were also used widely in the American Civil War (1861–65). The 1861 Springfield Rifled Musket

was used to devastating effect by Union soldiers. The guns fired three shots per minute and, in the hands of skilled marksmen, could consistently hit targets within 500 yards (457m).

AFTER >>

The percussion cap rendered all other ignition systems obsolete. It simplified the loading and firing process and made the revolver a viable proposition. It also paved the way for the development of the self-contained metallic cartridge and breech-loading firearms.

- **MAYNARD'S TAPE PRIMER** was one of the few percussion variations to enjoy a period of success, but it was flimsy and susceptible to damage compared to the copper cap.



Tape primer

- **THE REVOLVER** became a truly practical proposition. Early revolvers required a system to cover the pan to prevent the priming powder from falling out when the cylinder rotated. The cover also had to be moved when each chamber in the cylinder was in a firing position. Percussion caps solved these problems, allowing revolvers to be produced en masse.

- **BREECH-LOADING FIREARMS** such as the Dreyse needle-fire rifle (see pp.108–09) were developed. These used combustible cartridges in conjunction with separate percussion-cap ignition.

- **SELF-CONTAINED METALLIC CARTRIDGES** evolved using the percussion cap. Guns could be reloaded by merely opening the weapon's breech end, loading the cartridge, closing the breech, and cocking the weapon.



EARLY METALLIC CARTRIDGE

